Xiaoyu Peng

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EDUCATION

- **Ph.D. student in Electrical Engineering (Advisor: Prof. Feng Liu)** 2023 – Present
  Department of Electrical Engineering, Tsinghua University, China
- **B.Sc. in Electrical Engineering (Advisor: Prof. Feng Liu)** 2019 – 2023
  Department of Electrical Engineering, Tsinghua University, China

RESEARCH AND ENGINEERING

I work at the intersection of power-system dynamics and control theory, focusing on stability analysis, control, and the fundamental limitations of inverter-dominated power systems. A central theme of my research is to identify structural constraints that limit achievable stability, dynamic performance, and distributed stability certification, while developing physically interpretable and scalable methods for analysis and control. My work connects these theoretical insights with device-level assessment, control design, and practical power-system applications.

- **Power-System Stability under Angle–Voltage Coupling: Analysis, Control, and Fundamental Limitations** [1], [2], [3], [9], [11].

  At the power-system level, my work investigates how phase-angle and voltage-magnitude interactions affect system stability and dynamic performance. It includes revealing the damping-redistribution mechanism of angle–voltage coupling, formulating coupling-aware stability certificates as grid codes, developing implementation-oriented dynamic-decoupling control, and establishing Bode-type fundamental limits on achievable closed-loop performance.

- **Passivity-Based Distributed Stability Analysis: Compositionality, Visualization, and Fundamental Limitations** [1], [10], [12].

  At the control-theory level, my work develops passivity-based methods grounded in power-system dynamics for scalable distributed stability assessment. It establishes compositional criteria translating system-level stability and performance requirements into node-level constraints, geometric tools for interpreting passivization feasibility and margins, and structural limitations that characterize the applicability boundary of passivity-based distributed stability analysis.

- **Production-Oriented Software for Online Power-System Stability Assessment**.

  At the engineering level, I contribute to the architecture and development of online stability assessment software for practical power grid deployments. The framework converts online operating conditions into executable electromagnetic transient simulation models with automated consistency checking, parameter validation, and error correction; invokes existing simulators; and performs trajectory-based evaluations of both system-level stability and device-level active-support capability. The work also includes distributed simulation server scheduling, database/API/front-end integration, and deployment in utility environments.

PUBLICATIONS

I am the author/coauthor of more than 10 peer-reviewed technical papers and holds more than 10 issued or pending patents.

First-Author Publications

Peer-Reviewed Works

[1] **X. Peng**, C. Fu, Z. Li, X. Ru, Z. Wang, and F. Liu, "Compositional grid codes with guarantee on both stability and dynamic performance," *IEEE Transactions on Power Systems*, vol. 41, no. 3, pp. 2398–2401, 2026. DOI: 10.1109/tpwrs.2026.3651937

[2] **X. Peng**, Z. Li, C. Fu, P. Yang, Z. Wang, and F. Liu, “Stability degradation induced by angle–voltage coupling in power systems: Bode-type fundamental performance limitation analysis,” *IEEE Transactions on Automation Science and Engineering*, vol. 23, pp. 6915–6928, 2026. DOI: 10.1109/tase.2026.3676916
[3] **X. Peng**, C. Fu, P. Yang, X. Ru, and F. Liu, “Impact of angle-voltage coupling on small-signal stability of power systems: A damping perspective,” *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 73, no. 1, pp. 707–720, 2025. DOI: 10.1109/tcsi.2025.3620377
[4] **X. Peng**, F. Liu, P. Yang, B. Tan, P. Gao, and Z. Wang, “Measuring short-term voltage stability of power systems dominated by inverter-based resources, part ii: Port-wise generalized voltage damping index,” *Journal of Modern Power Systems and Clean Energy*, vol. 14, no. 1, pp. 82–94, 2025. DOI: 10.35833/mpce.2024.001298
[5] **X. Peng**, F. Liu, P. Yang, P. Yu, K. Luo, and Z. Wang, “Measuring short-term voltage stability of power systems dominated by inverter-based resources, part i: System-wise generalized voltage damping index,” *Journal of Modern Power Systems and Clean Energy*, vol. 13, no. 6, pp. 1871–1883, 2025. DOI: 10.35833/mpce.2024.001297
[6] **X. Peng**, Y. Shen, Q. Lu, and C. Shen, “Robust Var-Voltage Control Partitioning Method for Power Grid Considering Wind Power Uncertainty,” *Power System Technology (in Chinese)*, vol. 47, no. 10, 2023. DOI: 10.13335/j.1000-3673.pst.2022.2083
[7] **X. Peng**, Z. Li, K. Luo, T. Wang, S. Ma, and F. Liu, “Angle-voltage coupling can induce fundamental limits on voltage stability,” in *2025 Advanced Electrical and Energy Systems Conference*, 2025. DOI: 10.1109/aees66649.2025.11332447
[8] **X. Peng**, X. Ru, J. Zhang, T. Jiang, X. Chen, and F. Liu, “On the damping redistribution mechanism of angle-voltage coupling dynamics in power systems with hybrid gfm-gfl devices,” in *2025 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*, 2025. DOI: 10.1109/isgteurope64741.2025.11305652

Ongoing Works

[9] **X. Peng**, Z. Li, and F. Liu, *Angle-voltage dynamic decoupling control for stability enhancement of grid-forming inverters*, 2026.
[10] **X. Peng**, Z. Li, X. Ru, X. Chen, and F. Liu, *Relative-degree wall restricts passivity-based stability analysis in inverter-dominant grids*, 2026.
[11] **X. Peng** and F. Liu, *Recovering power-system synchronization: Conservation and performance tradeoffs in complex-frequency representation*, 2026.
[12] **X. Peng**, X. Ru, Z. Li, J. Zhang, X. Chen, and F. Liu, *Positive damping region: A graphic tool for passivization analysis with passivity index*, 2026. arXiv: 2601.10475.

Co-authored Publications

[13] Z. Li, **X. Peng**, X. Ru, Z. Wang, J. Zhang, Y. Liu, and F. Liu, *When can phasor-domain device models be trusted for electromechanical stability analysis of grid-forming converter-dominated microgrids?* 2026. arXiv: 2606.08082.
[14] X. Ru, C. Fu, Z. Li, **X. Peng**, and F. Liu, “Toward less conservative distributed stability analysis of power systems via matrix-valued differential passivity indices,” *iEnergy*, 2026.
[15] X. Ru, **X. Peng**, P. Yu, Z. Wang, P. Yang, and F. Liu, *Matrix-valued passivity indices: Foundations, properties, and stability implications*, 2026. arXiv: 2601.04796.
[16] Z. Sun, X. He, S. Jiang, **X. Peng**, J. Zhang, and H. Geng, “Quantifying power disturbance tolerance to guarantee voltage security in dc microgrids,” *IEEE Transactions on Smart Grid*, vol. 17, no. 4, pp. 3613–3616, 2026. DOI: 10.1109/tsg.2026.3684774
[17] Z. Sun, S. Jiang, **X. Peng**, X. Zhu, X. He, and H. Geng, *Decentralized frequency-domain conditions for d-stability with application to dc microgrids*, 2026. arXiv: 2605.13529.
[18] K. Kang, **X. Peng**, K. Luo, X. Ru, and F. Liu, “Controlled evolution-based day-ahead robust dispatch considering frequency security with frequency regulation loads and curtailable loads,” in *IEEE Power Engineering Society General Meeting*, 2025.
[19] Z. Li, X. Ru, **X. Peng**, B. Tan, P. Gao, and F. Liu, “How passivity of bus dynamics influences power system stability: A sensitivity analysis,” in *9th IEEE Conference on Energy Internet and Energy System Integration (IEEE EI2 2025)*, 2025. DOI: 10.1109/ei268505.2025.11425330
[20] Z. Li, B. Zeng, **X. Peng**, M. Wang, X. Ru, Y. Wang, and F. Liu, “Distributed stability analysis for power systems with grid-following and grid-forming devices,” in *4th Energy Conversion and Economics Annual Forum (ECE Forum 2024)*, vol. 2024, 2025, pp. 1463–1470. DOI: 10.1049/icp.2025.0735
[21] P. Yang, **X. Peng**, X. Ru, H. Geng, and F. Liu, *Compositional and equilibrium-free conditions for power system stability–part i: Theory*, 2025. arXiv: 2506.11406.
[22] P. Yang, Y. Su, **X. Peng**, H. Geng, and F. Liu, *Compositional and equilibrium-free conditions for power system stability–part ii: Method and application*, 2025. arXiv: 2506.11411.
[23] S. Wu, F. Liu, P. Yu, X. Ru, **X. Peng**, X. Zhou, and Y. Wu, “Unfolding the effects of energy storages on system frequency response: Equivalence principle and trajectory decomposition,” in *2023 IEEE 6th*

International Electrical and Energy Conference (CIEEC), 2023, pp. 1226–1231. DOI: 10.1109/cieec58067.2023.10167002

AWARDS

- **Excellent Comprehensive Scholarship (First Class, Graduate Student)** 2025
  Awarded by Tsinghua University
- **Outstanding Undergraduate Thesis Award** 2023
  Awarded by Tsinghua University
- **Excellent Comprehensive Scholarship (Undergraduate Student)** 2022
  Awarded by Tsinghua University
- **Excellent Comprehensive Scholarship (Undergraduate Student)** 2021
  Awarded by Tsinghua University

INDUSTRIAL COLLABORATION PROJECTS

- Research of Key Technologies of Power System Stability and Security with Considering Multiple Uncertainties, *National Science and Technology Major Project* 2024 – Present
  Role: research and software development on the electromagnetic-transient-simulation-based online stability evaluation for industrial deployments; project contact and coordination.
- Research on Fundamental Theory and Key Technologies for Security Networking Conditions in New Power Systems, *Science and Technology Project of State Grid Corporation of China* 2023 – 2026
  Role: research on the mechanism and quantification of voltage strength in grids; project contact and coordination.
- Research on Evolution and Mechanism of Stability Characteristics in Ultra-Large Provincial Power Systems, *Science and Technology Project of China Southern Power Grid* 2023 – 2025
  Role: research on the dynamic mechanism of how the coupling between phase-angle and voltage dynamics influences the stability of inverter-dominated power systems; project contact and coordination.
- Research and Application of Key Technologies of Intelligent Cooperative Control of Large-Scale Energy Storage System Cluster, *National Key Research and Development Program* 2022 – 2024
  Role: research and software development on the measurement-based quantification of voltage stability support capability of energy storage systems.
- Research on Control of Power Systems with High Proportions of Renewable Energy and Power Electronics, *Science and Technology Project of State Grid Corporation of China* 2022 – 2024
  Role: investigation on the impact of energy storage systems on dynamics and stability of power systems with high proportions of renewable energy and power electronics.

TEACHING EXPERIENCE

- **Teaching Assistant**, Signal, System and Control (Prof. Feng Liu and Prof. Jin Lin), Weiyang College, Tsinghua University. 2023 – 2024
- **Teaching Assistant**, Principle of Automatic Control (Prof. Jin Lin and Prof. Feng Liu), Department of Electrical Engineering, Tsinghua University. 2022 – 2025

ACADEMIC SERVICE

- **Peer Reviewer**, *IEEE Transactions on Power Systems*, *IEEE Transactions on Smart Grid*, *IEEE Transactions on Sustainable Energy*, and *IEEE Power Engineering Letters*. 2024 – Present

ADDITIONAL INFORMATION

- **Languages:** Chinese (native), English (fluent).
- **Hobbys:** cycling, hiking and bird watching.